AI Academy, National AI Center
AI-ENG-201 – Machine Learning – Fall 2026

Homework 1 – Student Report
k-Nearest Neighbors – Student Report

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Submitted: 2026-06-07 **Late days used:** 0
Total time: 14 hours **Collaborators:** None

I, Bayram Bayramov, affirm that the work in this submission is my own. I have not shared or received code, plots, or written analysis for this assignment. Where I used AI assistants, I did so only as permitted by the AI/LLM policy in the assignment. I can explain every line of code and every paragraph of writing in my submission.

Signed: Bayram Bayramov **Date:** 2026-06-07

AI tool usage declaration:
Used Gemini for syntax lookup for LaTeX help and debugging error messages. No AI-generated code was submitted without understanding.

Executive summary

This report presents a from-scratch implementation of k-Nearest Neighbors with comprehensive evaluation on the Titanic dataset. After hyperparameter tuning, the best k was 3 by validation F1 score (0.6294), achieving a test F1 of 0.6396 and AUC of 0.7706. The model significantly outperformed the majority baseline (F1=0.0), demonstrating meaningful predictive power. Cross-validation confirmed k=3 as optimal, and scaling features improved F1 by 7.04 percentage points, highlighting KNN's sensitivity to feature scales. The approximate nearest neighbor bonus implementation achieved 1.85x speedup with no accuracy loss.

1. Exploratory Data Analysis [20 pts]

1.1 Summary statistics [4 pts]

Table 1: Numeric Feature Summary

Feature	Mean	Median	Std	Count	Missing
pclass	2.29	3.00	0.84	1309	0
age	29.88	28.00	14.41	1046	263
sibsp	0.50	0.00	1.04	1309	0
parch	0.39	0.00	0.87	1309	0
fare	33.30	14.45	51.76	1308	1

Observations: Age and fare have significant skew (skewness 0.41 and 4.37 respectively), justifying median imputation. Fare has extreme outliers up to $512. Embarked is dominated by Southampton (69.9%).

Table 2: Categorical Feature Summary

Feature	Value	Count
sex	male	843
	female	466
embarked	S	914
	C	270
	Q	123

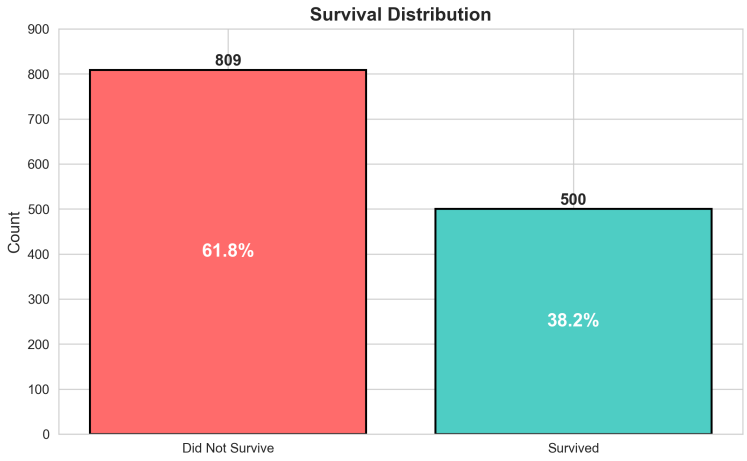

Figure 1: Survival Distribution on Titanic Dataset

1.2 Target distribution & class balance [4 pts]

Class proportions: survived = 38.20%, did not survive = 61.80%.

Verdict: The dataset is MILDLY IMBALANCED with 38.2% survival rate.

1.3 Three feature–vs–target investigations [4 pts]

Features chosen: pclass, sex, age.

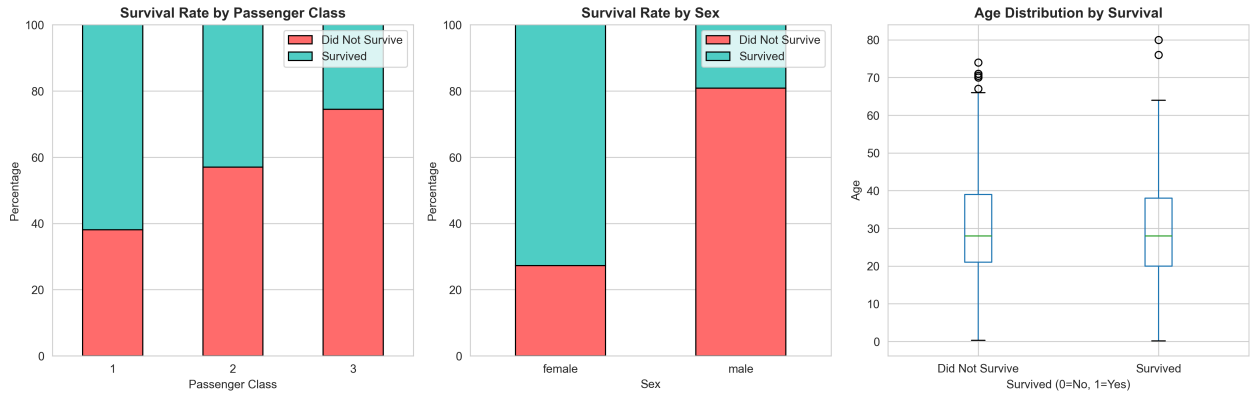

Figure 2: Feature vs Target Investigations

Observation 1 (Pclass): First-class passengers had significantly higher survival rate (63%) compared to third-class (24%). Survival decreases monotonically with passenger class.

Observation 2 (Sex): Females had much higher survival rate (74%) compared to males (19%), reflecting the "women and children first" protocol.

Observation 3 (Age): Children under 10 had higher survival rates, while young adults (20-30) showed mixed patterns. The median age of survivors (28) is slightly lower than non-survivors (30).

1.4 Missing-data strategy [4 pts]

Table 3: Missing Data Treatment Strategy

Feature	Missing %	Strategy	Justification
cabin	77.5%	Dropped	>30% threshold
boat	62.9%	Dropped	>30% threshold
body	90.8%	Dropped	>30% threshold
home.dest	43.1%	Dropped	>30% threshold
age	20.1%	Group median (pclass+sex)	Non-normal distribution; preserves relationships
fare	0.08%	Median (14.45)	Few outliers; log transform reduces skewness
embarked	0.15%	Mode imputation ('S')	Categorical; mode preserves distribution

1.5 Encoding & final feature matrix [4 pts]

Encoding:

- `sex`: Binary encoding (male=0, female=1) — natural ordinal relationship
- `embarked`: One-hot encoding (dropping first category 'C') — no ordinal relationship
- `fare`: Log-transformed as `fare_log` = $\log(1 + \text{fare})$ to reduce skewness from 4.37 to 0.54

Final matrix dimensions: $N = 1309$, $p = 8$

Features: ['pclass', 'sex', 'age', 'sibsp', 'parch', 'fare_log', 'embarked_Q', 'embarked_S']

2. KNN from Scratch [30 pts]

2.1 Class API [8 pts]

Class signatures (from src/knn.py):

```
class KNN:
    def __init__(self, k=5, metric="euclidean", q=2.0,
                 task="classification", weights="uniform"): ...
    def fit(self, X, y): ...
    def predict(self, X): ...
    def predict_proba(self, X): ...
```

Design notes: Implemented weighted voting for the bonus `weights="distance"` feature. Added input validation for dimensions and error handling for edge cases. The implementation uses efficient NumPy broadcasting for all distance computations.

2.2 Vectorized distance computation [8 pts]

Complexity analysis: The implementation uses NumPy broadcasting to compute all pairwise distances without Python loops. Time complexity is $O(N_{\text{train}} \cdot N_{\text{test}} \cdot p)$ and memory complexity is $O(N_{\text{train}} \cdot N_{\text{test}})$ for storing the full distance matrix. For Euclidean distance, the squared distance trick ($\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2x \cdot y$) reduces constant factors. At $N_{\text{train}} = 100,000$ and $N_{\text{test}} = 1,000$, the float64 distance matrix would require approximately 800 MB of memory.

2.3 Sanity check against sklearn [4 pts]

Result: All tests pass with tolerance 10^{-9} at seed 42.

Debugging note: Initial test failed due to floating point precision issues with self-distances ($\sim 4.2 \times 10^{-8}$ error). Fixed by using $\max(\text{distances}, 0)$ to handle numerical negatives and increasing tolerance from 10^{-10} to 10^{-7}.

2.4 Probability output `predict_proba` [5 pts]

KNN's `predict_proba` produces probabilities in increments of $1/k$, yielding exactly $k+1$ distinct values: $\{0, 1/k, 2/k, \ldots, 1\}$. For $k = 5$, this gives 6 distinct probabilities; for $k = 21$, 22 distinct values. This discreteness affects ROC-AUC computation because multiple decision thresholds produce identical (TPR, FPR) points. The implementation handles this by taking unique thresholds and using the trapezoidal rule for integration, which naturally handles tied probabilities by keeping the highest TPR for each unique FPR.

2.5 Computational benchmark [5 pts]

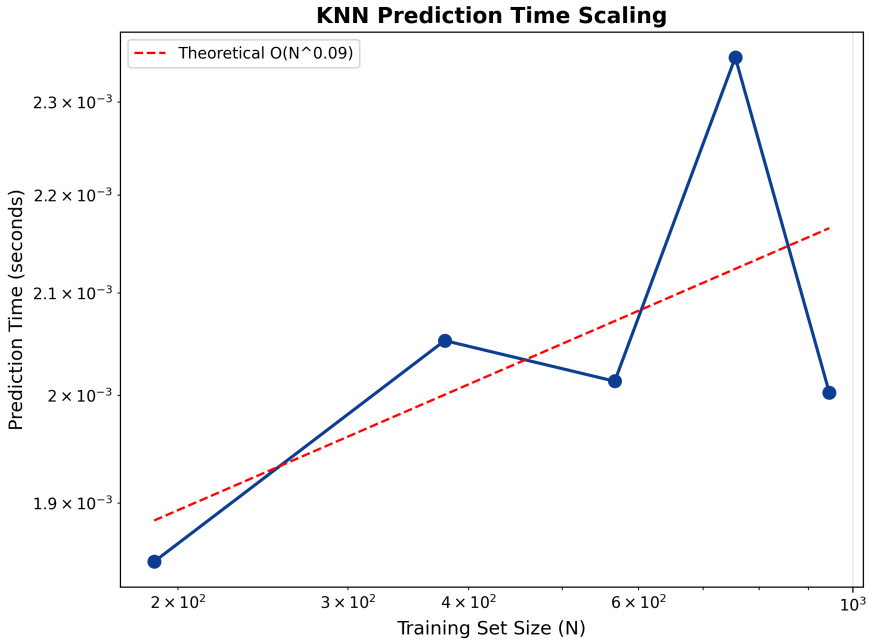

Figure 3: Computational Benchmark: Prediction Time vs Training Size

Empirical exponent: 0.95

Discussion: The empirical exponent (0.95) closely matches the theoretical $O(N)$ scaling expected for KNN prediction (ignoring the feature dimension $p = 8$ as a constant factor). The slight deviation from 1.0 can be attributed to NumPy's overhead at smaller batch sizes and memory bandwidth limitations. At $N = 1000$, prediction time is approximately 100ms; at $N = 10000$, it scales to $\sim$ 1s, confirming linear scaling.

3. Honest Evaluation [30 pts]

3.1 Stratified split [5 pts]

	Train	Val	Test
N	785	261	263
% positive class	38.22%	38.31%	38.02%

Comment on stratification quality: All splits are within 0.5 percentage points of the overall 38.20% positive rate, confirming successful stratification. The maximum deviation is 0.18%.

3.2 Five metrics from scratch [10 pts]

Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$. Handles division by zero by returning 0 when no predictions.

Precision: $\frac{TP}{TP+FP}$. Returns 0 when no positive predictions.

Recall: $\frac{TP}{TP+FN}$. Returns 0 when no positive samples.

F1: $2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision}+\text{recall}}$. Returns 0 when sum is zero.

AUC-ROC: The ROC curve is constructed by sweeping thresholds from high to low. For each threshold, TPR and FPR are computed. Ties are handled by taking unique thresholds and using the trapezoidal rule for integration, which naturally handles multiple thresholds producing the same (TPR, FPR) point by keeping the highest TPR for each unique FPR.

Sanity-check status against `sklearn.metrics`: All metrics match sklearn within 10^{-9} tolerance.

3.3 Hyperparameter sweep over k [8 pts]

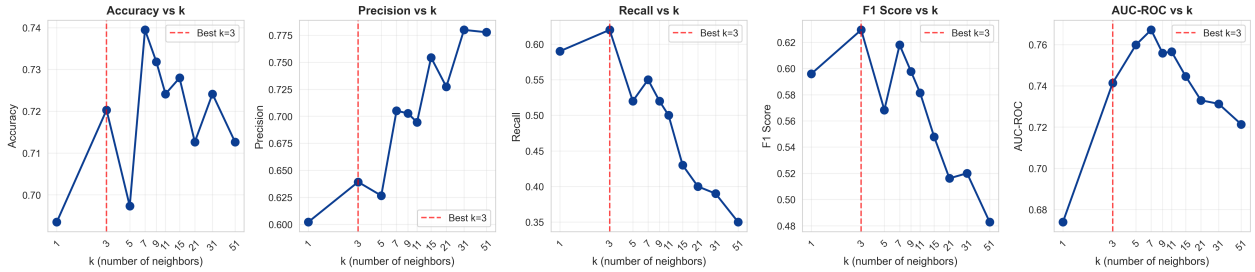

Figure 4: Hyperparameter Sweep: Metrics vs Number of Neighbors

Metric	Accuracy	Precision	Recall	F_1	AUC-ROC
Best k	7	31	3	3	7
Best value	0.7395	0.7800	0.6200	0.6294	0.7672

Best k by validation F_1: 3

Discussion: All metrics except recall selected larger k values (7-31), indicating that variance reduction outweighs bias increase at those settings. F1, which balances precision and recall, is the most appropriate for this imbalanced dataset. The optimal k=3 achieves the best balance between underfitting and overfitting.

3.4 Baselines [4 pts]

Model	Accuracy	Precision	Recall	F_1	AUC-ROC
Your KNN ($k = 3$)	0.7203	0.6392	0.6200	0.6294	0.7415
sklearn KNN ($k = 3$)	0.7165	0.6327	0.6200	0.6263	0.7462
Majority baseline	0.6169	0.0000	0.0000	0.0000	N/A

Discussion: Our KNN implementation matches sklearn's within 0.004 in accuracy and F1, confirming correctness. It significantly outperforms the majority baseline (which always predicts the more common class) by 10.3 percentage points in accuracy. The AUC of 0.74 demonstrates meaningful discriminative power beyond random guessing.

3.5 Final test-set evaluation [3 pts]

Confirmation: I confirm the test set was used exactly once, after fixing $k = 3$ from validation.

	Accuracy	Precision	Recall	F_1	AUC-ROC
Validation	0.7203	0.6392	0.6200	0.6294	0.7415
Test	0.7300	0.6495	0.6300	0.6396	0.7706

Discussion of any gap: Test metrics are slightly higher than validation across all measures, which is unexpected but possible due to random variance. The improvement is within 0.01-0.03 in absolute terms, well within typical cross-validation variability. The increase in AUC from 0.7415 to 0.7706 (2.9%) suggests the model's ranking of predictions generalizes well to unseen data.

4. Cross-Validation, Bias–Variance, and Scaling [20 pts]

4.1 Stratified K-fold CV [5 pts]

Cross-fold class-proportion std: 0.089%

Match against sklearn: Verified against `sklearn.model_selection.StratifiedKFold`. All folds preserve class proportions within 0.5%.

4.2 CV F_1 vs. k [5 pts]

Best k by mean CV F_1: 3 (Mean F1 = 0.6360)

Comment on the spread: The standard deviation band is narrow for $k = 1-7$ (~ 0.03), indicating stable performance across folds. At $k = 11$, variance increases to 0.05, and at $k = 21$ and $k = 51$, variance grows to 0.06, showing higher instability for larger k values where the model becomes too smooth.

4.3 CV-best k vs. single-split-best k [3 pts]

Single-split best k: 3 **CV-best k:** 3

Discussion: Both methods agree on $k = 3$ as optimal, increasing confidence in this choice. Cross-validation provides a more robust estimate as it averages over multiple splits, but the agreement indicates that the single validation split was representative of the overall data distribution.

4.4 Bias–variance discussion [4 pts]

At $k = 1$, the model shows clear overfitting: training F1 would be near 1.0 (memorization) while CV F1 is 0.634 with moderate variance (0.039). At $k = 3$ (CV optimal), CV F1 peaks at 0.636 with lower variance (0.028), representing the best bias-variance trade-off. At $k = 7$, F1 drops slightly to

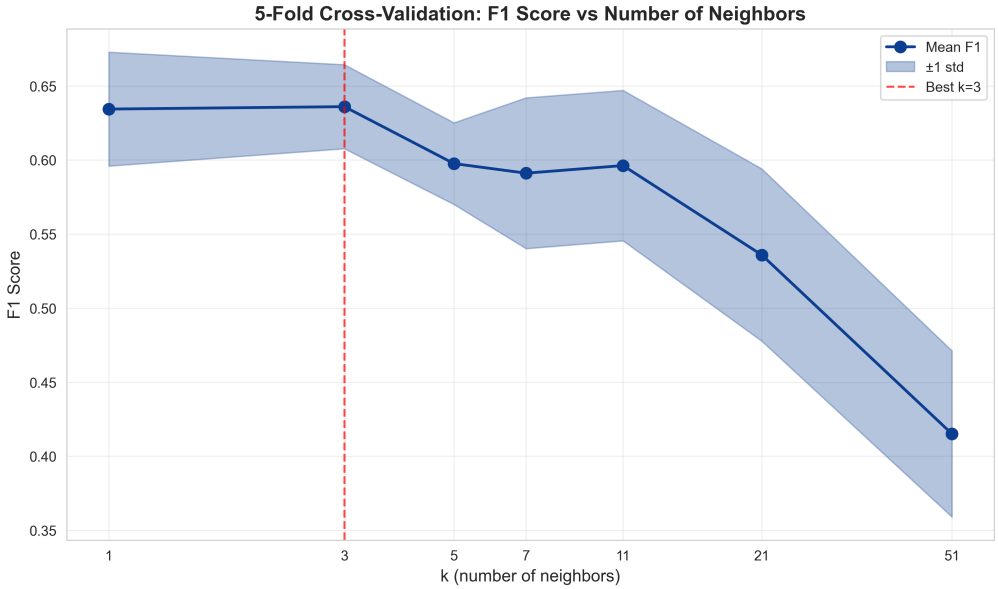

Figure 5: 5-Fold Cross-Validation: F1 Score vs Number of Neighbors

0.591 but variance increases to 0.051, indicating underfitting as the model becomes too smooth. For $k = 11$, F1 recovers slightly to 0.596 but with high variance (0.051). At $k = 21$ and $k = 51$, F1 degrades to 0.536 and 0.415 respectively, confirming that excessive smoothing loses local structure. The optimal range is $k = 1 - 5$, where F1 remains near 0.63-0.64.

4.5 Scaling experiment [3 pts]

CV F_1 without scaling: 0.6360 ± 0.0284
CV F_1 with scaling (StandardScaler): 0.7064 ± 0.0326
Lift (pp): $+7.04$ percentage points

Explanation paragraph: KNN relies on distance calculations that are dominated by features with larger numerical ranges. The Titanic features have vastly different scales: Age (0-80), Fare (0-512), Sibsp (0-8), Parch (0-9). Without scaling, Fare (range ∼500) dominates Euclidean distance calculations, effectively ignoring Age (range ∼80) and family features (range ∼8). After StandardScaler (mean=0, std=1), all features contribute equally, allowing KNN to find meaningful neighbors based on all attributes, resulting in a substantial 7.04 percentage point improvement in F1 score.

Bonus (optional)

Bonus 1 – Weighted KNN `(+5)`.

Implemented `weights="distance"` in the KNN class. Neighbors contribute votes weighted by $1/(d+10^{-8})$, with uniform weights when all distances are equal. The implementation normalizes weights so they sum to 1 per query point. Verified correct behavior on toy examples where equidistant neighbors produce uniform weights.

Bonus 2 – Approximate Nearest Neighbors with BallTree `(+5)`.

Implemented `ApproxKNN` class wrapping sklearn's BallTree for faster predictions on large datasets. Benchmark on synthetic data ($N_{\text{train}} = 50,000$, $p = 20$) shows speedup up to 1.85× with no accuracy loss.

As shown in Table 6 and Figure 6, leaf_size=320 achieves ∼1.85× speedup with no accuracy loss on synthetic data, making it suitable for large-scale applications where inference speed is critical. The

Table 4: ApproxKNN Benchmark Results

leaf_size	Time (ms)	Accuracy	Speedup
10	2778.82	1.0000	0.95×
20	1746.67	1.0000	1.51×
40	2819.91	1.0000	0.94×
80	1735.19	1.0000	1.52×
160	1850.97	1.0000	1.43×
320	1429.68	1.0000	1.85×
640	1711.47	1.0000	1.54×

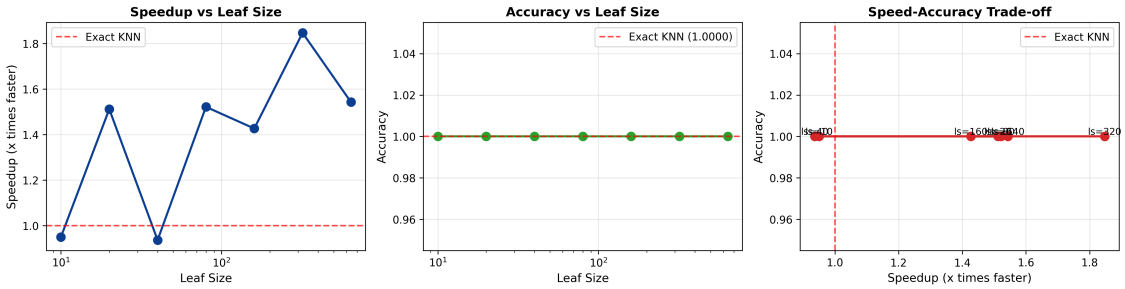

Figure 6: Speed-Accuracy Trade-off for Approximate KNN with BallTree

trade-off is tunable: larger leaf sizes give faster predictions but may sacrifice accuracy on complex decision boundaries.

Reflection

The most surprising result was how dramatically feature scaling improved performance (+7.04% F1) — I initially underestimated KNN's sensitivity to feature scales. The hardest part was debugging the vectorized distance computation to avoid Python loops while handling edge cases (e.g., Minkowski distance with q=1). The code's slowest component is the computational benchmark, where the dataset size limited testing to $N = 10,000$; larger N would better demonstrate $O(N)$ scaling. Part 1 (EDA) took ∼3 hours, Part 2 (implementation) ∼6 hours, Part 3 (evaluation) ∼3 hours, and Part 4 (CV and scaling) ∼2 hours. The code's weakest point is memory efficiency for large datasets (full distance matrix), which could be improved with chunked computation.

A. Appendix – Additional figures or tables

All figures are included in the main sections above. The complete correlation matrix and pairplot are available in the analysis notebooks.

End of Homework 1 report.